PRIDE DBAlign

MANUAL

Multi-GNSS satellite product

Day-Boundary discontinuity Alignment

by

PRIDELab

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Dedicated to those
who are devoted to high-precision GNSS

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1 Introduction

1.1 Acknowledgements

PRIDE DBAlign is used for the day-boundary discontinuity alignment of GNSS precise orbit/clock/bias products, developed by the PRIDE team (http://pride.apm.ac.cn). We make this software package open source with the goal of promoting the implementation of United Nations (UN) International Committee on GNSS (ICG) 2025 recommendation, "Continuous GNSS Time Transfer across Day Boundaries using IGS Products" (https://www.unoosa.org/documents/pdf/icg/2025/ICG-19/ICG-19_WG-D_Recommendation_B_.pdf). In particular, PRIDE DBAlign is able to align precise satellite products across day boundaries to keep ambiguities continuous beyond the midnight epochs. As a result, GNSS positioning and time/frequency transfer spanning days could be improved by avoiding day-boundary “jumps”. PRIDE DBAlign utilizes several library functions, header files, and table files from PRIDE PPP-AR (https://github.com/PrideLab/PRIDE-PPPAR), another open-source software developed by the same team. The development of PRIDE DBAlign software is funded by the National Natural Science Foundation of China (Grant No. U25D8020) and supported by Sub-Commission 4.2 of the International Association of Geodesy (IAG), the Wuhan Combination Center (WCC, https://igs.org/wg/wcc/), and the Bias & Ambiguity Resolution Committee (BAR, https://igs.org/wg/bias-and-ambiguity-resolution/) of the International GNSS Service (IGS).

The open-source software PRIDE DBAlign can be downloaded at https://github.com/PrideLab/PRIDE-DBAlign. For the latest updates regarding support, and frequently asked questions (FAQs), please visit http://pride.apm.ac.cn/. The copyright of this software package is protected under the GNU General Public License (version 3). Relevant publications are

- Lin J, Geng J, Zhang Q (2025) Aligning GPS/Galileo/BDS satellite integer clock products across day boundaries for continuous time and frequency transfer. J Geod 99, 35. https://doi.org/10.1007/s00190-025-01955-5
- Wen Q, Geng J, Deng Y, Zhang Y (2025) Validating the IGS products in mitigating day-boundary discontinuities of kinematic positioning and time transfer. GPS Solut 29, 170. https://doi.org/10.1007/s10291-025-01929-2
- Geng J, Wang Y, Wen Q, Lin J, Tagliaferro G (2026) Quantifying all-frequency day-boundary discontinuities for GPS/Galileo/BDS satellite products. (*under review*)

1.2 Major Features

PRIDE DBAlign is capable of performing day-boundary alignment on multi-GNSS precise clock/bias products to restore their cross-day continuity. It can be applied to research involving cross-day precise positioning and timing, such as geodesy and time transfer. The main features of PRIDE DBAlign include:

1) Support for GPS, Galileo, and BDS-2/3;
2) Support for the alignment of precise satellite products on baseline frequencies (*e.g.*, GPS L1/L2);
3) Provide a DOCB(Discontinuity of orbit, clock, and bias) module in the Bias-SINEX

format for day-boundary alignment residuals;

4) Aligned precise products are consistent with the latest IGS conventions: the Bias-SINEX bias file format;
5) An additional plotting function is provided to display day-boundary alignment residuals, facilitating data analysis for researchers.

1.3 Contact Us

If you encounter any issues during the software usage, you can contact us by sending an email to report **bugs** and seek assistance.

Email: pride@apm.ac.cn

Website: http://pride.apm.ac.cn/

2 Version modifications

- 2026-09-18 (v1.0)
  - Release of PRIDE DBAlign v1.0

3 Program structure and algorithm

3.1 Program structure

The core program of the PRIDE DBAlign software is algirc. Its operational workflow is illustrated in Figure 3-1 and can be generally divided into four parts: variable definition and initialization, configuration and data reading, dual frequency precise product alignment, and output of log file and aligned products.

Variable definition and initialization. This part mainly handles the definition and initialization of variables, constants, and control parameters required for the program to run, including the satellite list, clock offset, orbit, code/phase biases, day-boundary discontinuities, and corrections. In addition, based on different GNSS systems, it initializes the frequencies, frequency ratios, combination coefficients, and wavelength parameters required for wide-lane, narrow-lane, and ionosphere-free combinations. This provides the foundational parameters for subsequent day-boundary discontinuity estimation and product alignment.

Configuration and product reading. The program first reads the configuration file and saves the product directory, SP3 precise orbit files, CLK precise clock offset files, BIA code/phase bias files, satellite list, and related control options into their corresponding variables. Subsequently, it sequentially reads the precise clock offset, precise orbit, and code/phase bias products of two adjacent days to calculate the day-boundary discontinuities of various precise products. For precise clock offset products, if the preceding day's CLK file contains the 24:00:00 epoch (i.e., the 00:00:00 epoch of the target day), it calculates the clock offset day-boundary discontinuity directly using the clock offsets of the same epoch in the two adjacent files. Otherwise, a least-squares linear extrapolation is performed using the last ten clock epochs at the end of the preceding day to extrapolate the clock offset for the preceding day at 24:00:00, provided that the extrapolation interval does not exceed 300 seconds. The extrapolated clock offset is then used to calculate the day-boundary discontinuity. When products are output, the program writes an additional CLK file for the preceding day with the “_predicted” suffix, preserving the original content and appending the extrapolated clock records after the last epoch of the original file; the target day's 00:00:00 epoch must already exist in its input CLK file and is never synthesized. For precise orbit products, the orbit files of both adjacent days are required. If the orbit products of the two adjacent days contain the required day-boundary epoch, the program directly calculates the orbit day-boundary discontinuity using the orbit positions at the boundary. Otherwise, the orbit position at the boundary is extrapolated using trigonometric-polynomial fitting and is then used to calculate the orbit day-boundary discontinuity. When orbit extrapolation is performed, the program writes an additional SP3 file with the “_predicted” suffix, preserving the original content and appending the extrapolated orbit records after the last epoch of the original file. The program projects the orbit difference at the day boundary into the satellite's radial direction and converts it into an equivalent time correction. For BIA code/phase bias products, the code and phase biases of corresponding satellites and frequencies are read across adjacent days. It calculates the day-boundary discontinuities of combined quantities such as wide-lane phase bias, narrow-lane phase bias, geometry-free bias, and code bias, providing constraints for the subsequent alignment of the integer clock and phase biases.

Dual-frequency precise clock/bias product alignment. This section is the core of the algirc program, primarily used to eliminate the day-boundary discontinuities in dual-frequency precise

clock/bias products of adjacent days. First, the program calculates the wide-lane phase bias day-boundary differences based on the BIA products of adjacent days and estimates the wide-lane datum correction for each satellite system to remove integer cycle jumps. Next, combining the precise clock offset day-boundary differences and the radial orbit day-boundary differences, it calculates the clock offset datum correction and constructs the day-boundary discontinuity for the Integer Recovery Clock (IRC). The program then performs an iterative estimation on the narrow-lane phase bias and IRC day-boundary differences to determine the narrow-lane datum correction and eliminate the integer narrow-lane cycle component. Finally, the wide-lane and narrow-lane corrections are converted into phase bias corrections at each original frequency, simultaneously achieving consistent correction results for clock offsets, and phase biases. Through these steps, a unified day-boundary alignment of dual-frequency precise clock offset and phase bias products is achieved.

Log file and aligned product output. After completing the dual-frequency precise product alignment, the program generates an operation log file recording the input files, processing parameters, day-boundary discontinuity calculation results, as well as clock offset, orbit, and bias corrections. This is used to verify the product alignment process and the quality of the results. Subsequently, the program generates aligned CLK and BIA products with the "_aligned" suffix. For clock offset products, the corrected satellite clock offsets of the target day are written. If the preceding day's CLK file lacks 24:00:00, an additional '_predicted' CLK file for the preceding day is written as an unchanged copy with the extrapolated records appended after the last original epoch. When orbit extrapolation is performed, an additional '_predicted' SP3 file is generated with the extrapolated orbit records appended after the last original epoch. For BIA products, it primarily updates the phase biases at the corresponding frequencies and can output the DOCB (Discontinuity of orbit, clock, and bias) information as needed.

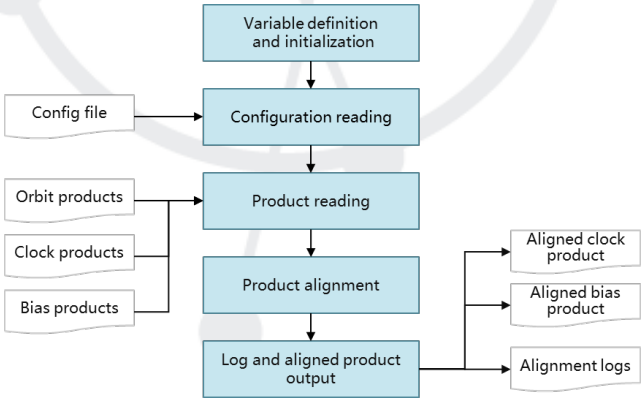


Figure 3-1 Flowchart of the algirc program

3.2 pdba command script

pdba (Product Day-Boundary Alignment) is the command-line driver script in PRIDE DBAlign for aligning GNSS precise products across a day boundary. Each invocation processes one target day and aligns its clock and bias products to the corresponding products of the preceding day. The processing date and preceding day product names are determined automatically from the input filenames. Users must ensure that the target day and preceding day products are placed in the same directory and that all command-line parameters are entered correctly.

The main functions of the current ***pdba*** script are described below.

- The ***main()*** function is the entry of the script, and its workflow is as follows:
  1. Parse the command-line arguments, including the clock, bias, SP3, configuration, GNSS system, write DOCB, and BDS-2/3 datum options.
  2. Check that the required clock, orbit and bias products exist and resolve their absolute paths.
  3. Calculate the preceding date and locate the corresponding preceding day clock, orbit and bias products in the same directory.
  4. Generate config_pdba_YYYYDDD in the product directory and update the product names, GNSS systems, and BDS-2/3 datum settings.
  5. Locate the algirc executable through the system PATH. Invoke algirc in the product directory to perform day-boundary alignment.
  6. Report the effective processing settings, input products, generated configuration, and completion status.
- The time-conversion functions include ***ydoy2mjd()***, ***ymd2mjd()***, ***mjd2ydoy()***, ***ydoy2ymd()***, ***gpswd2mjd()***, and ***mjd2gpswd()***:
  1. *ydoy2ymd()* converts a year and day of year into the corresponding year, month, and day.
  2. *ymd2mjd()* converts a year, month, and day into Modified Julian Date (MJD).
  3. *ydoy2mjd()* converts a year and day of year into MJD.
  4. *mjd2ydoy()* converts an MJD into the corresponding year and day of year.
  5. *gpswd2mjd()* converts a GPS week and day of week into MJD.
  6. *mjd2gpswd()* converts an MJD into the corresponding GPS week and day of week.
- The product-date processing functions include ***extract_product_date()***, ***calculate_previous_date()***, and ***previous_product_name()***:
  1. *extract_product_date()* detects and validates the date field in a product filename. It supports the long YYYYDDD[HHMM] convention and the legacy WWWWD GPS-week/day convention.
  2. *calculate_previous_date()* subtracts one day from the target date and calculates the preceding year, day of year, GPS week, and day of week.
  3. *previous_product_name()* replaces only the detected date field in the target filename to construct the expected preceding day product filename.
- The configuration-processing functions include ***set_config_value()*** and ***get_config_logical()***:
  1. *set_config_value()* updates a specified configuration item. If the item does not exist, it is appended to the generated configuration file.
  2. *get_config_logical()* reads and validates a logical configuration value such as YES or NO.
- The GNSS system functions include ***normalize_systems()***, ***filter_satellite_systems()***, and ***inspect_satellite_systems()***:
  1. *normalize_systems()* validates the value of -sys and normalizes the selected systems into the standard GEC order.
  2. *filter_satellite_systems()* comments out satellites belonging to systems that were not selected on the command line.
  3. *inspect_satellite_systems()* examines the generated configuration and determines the GNSS systems that will actually be processed.
- The output and error-handling functions include ***print_usage()***, ***print_version()***, ***error()***, ***warning()***, and ***die()***:
  1. *print_usage()* displays the command syntax, option descriptions, default values, processing rules, and examples.
  2. *print_version()* displays the current version of pdba.
  3. *error()* and *warning()* output diagnostic messages.
  4. *die()* outputs an error message and terminates the script.

4 Technical Aspects

4.1 User Requirements

4.1.1 System Requirements

PRIDE DBAlign is composed of CUI APs(Command User Interface Application Programs, CUI APs). The executable binary CUI APs included in the package support Linux and macOS. A series of tests are conducted on different operating systems with several gfortran versions. The test results are listed as in Table 4-1. Note that you can also try other Linux distribution and Fortran compiler, and tell us if you have any problems.

Fortran compiler, such as gfortran needs to be installed before installing PRIDE DBAlign.

Table 4-1 PRIDE DBAlign test results in different operating systems.

Platform version	gfortran version	Test result	Notes
Ubuntu20.04.5(x64)	11.4.0	pass	Test result is consistent with the reference
Ubuntu20.04.6(x64)	9.4.0	pass	Test result is consistent with the reference
CentOS 7 (x64)	4.8.5	pass	Test result is consistent with the reference
Debian 10(x64)	8.3.0	pass	Test result is consistent with the reference
MacOS 15.4.1	16.1.0	pass	Test result is consistent with the reference
MacOS 26.5.1	16.1.0	pass	Test result is consistent with the reference

4.1.2 License

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4.2 Installation Guide

4.2.1 Structures of PRIDE DBAlign

The structure and main files of the PRIDE DBAlign are listed in Table 4-2. We provide a complete set of cases in the “example/” folder for users to execute. To provide users with a better understanding of the software structure, further details are given below.

Table 4-2 PRIDE DBAlign structures

Directory / File	Directory / File	Instructions
bin/		**Executable program**
	algirc	Product alignment program
scripts/		**Scripts that facilitate data process**
	pdba.sh	Automatic processing Shell script
	pdba_plot.sh	Automatic plot Shell script
	plot-orbdbd.py	Plot heatmap of radial orbit DBDs
	plot-docb.py	Plot heatmap of DOCBs
src/		Source programs
	algirc/	Product alignment
	header/	Header files

	lib/	Library functions
	Makefile	Compile Command
table/		**Dependent table files**
	config_template	Configuration file template
	sat_parameters	Satellite parameters
	leap.sec	Leap seconds
install.sh		Installation script
CHANGELOG.md		Change log of the software
README.md		Software related information
LICENSE.txt		GPL3 protocol
doc/		**Documentation**
	logo	The logo printed on screen when installing successfully
	Manual	Manual of the software
example/		Examples
	wum0rap/	**Data of examples**
	test.sh	Test script

4.2.2 Installation and validation

Step 1: Make sure you have installed some essential programs in advance.

i.e. **bash, make, gfortran**

Step 2: Run script ***./install.sh*** to install the program automatically. (This script executes Makefile to build CUI APs and add CUI APs to system PATH (**~/.PRIDE_DBAlign_BIN**))

If you see the prompt shown in the picture below, the installation is completed normally.



Figure 4-1 successfully installed interface

Step 3: Input y/Y to run the example.

The script ***test.sh*** in example/ folder is used to validate the correctness and effectiveness of the installation and execution. After "test.sh" has been executed successfully, the WUM-rapid product day-boundary alignment example will be run. During data processing, progress and status information will be printed to the screen. After the run is complete, compare the generated result

files with the reference results to verify that the software has been installed correctly and is functioning properly.

4.3 File Specifications

4.3.1 Result Files

After the test script has been executed successfully, the result files are generated in the "example/wum0rap/2025" and "example/wum0rap/2026" directory, as shown in Figure 4-2. The 2025 results are used as an example in the following description.

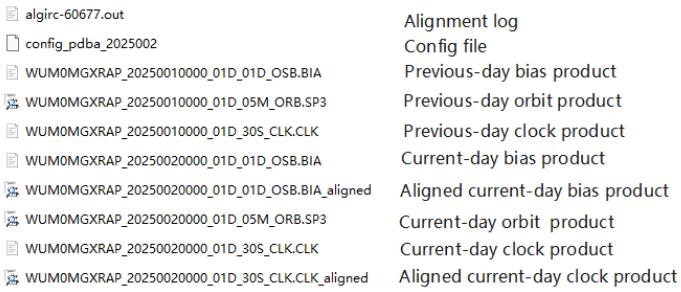


Figure 4-2 Result files after Product Alignment

algirc-MJD.out File

The alignment log records intermediate parameter estimates and product alignment residuals generated during the alignment procedure, thereby allowing users to assess the alignment performance. The log file comprises six sections: the file header; the availability of products at the 24:00 epoch, orbit and clock day-boundary discontinuities; dual frequency code bias and wide-lane day-boundary discontinuities; narrow-lane phase bias day-boundary discontinuities; and alignment corrections and residuals for code and phase biases.

The header format is illustrated below. It reports the executable name, execution time, working directory, product file names and configuration.

```
%%% INTEGER-RECOVERED CLOCK DAY-BOUNDARY DISCONTINUITY ANALYSIS FILE
%%% Executor:                    algirc
%%% Execute Time:                2026-10-09 20:48:09
%%% Working Directory:           /home/wyy/code/PRIDE-DBAlign/example/wum0rap/2025
%%% Input:                       /home/wyy/code/PRIDE-DBAlign/example/wum0rap/2025/config_pdba_2025002
%%% SP3 Files:                   WUM0MGXRAP_20250010000_01D_05M_ORB.SP3 WUM0MGXRAP_20250020000_01D_05M_ORB.SP3
%%% CLK Files:                   WUM0MGXRAP_20250010000_01D_30S_CLK.CLK WUM0MGXRAP_20250020000_01D_30S_CLK.CLK
%%% OSB Files:                   WUM0MGXRAP_20250010000_01D_01D_OSB.BIA WUM0MGXRAP_20250020000_01D_01D_OSB.BIA
%%% Write DOCB:                  YES
%%% Unify BDS-2/3:               NO
```

Figure 4-3 Header of the Alignment Log File

As illustrated below, this section reports the availability of products at the 24:00 epoch and the radial-orbit and clock day-boundary discontinuities in millimeters and picoseconds, respectively, together with the clock adjustment applied at the day boundary.

```
0. AVAILABILITY OF PRODUCTS AT 24:00 EPOCH
 ORBIT_24:00_EXIST: YES
 CLOCK_24:00_EXIST: YES

1. RESULTS OF RADIAL ORBIT CORRECTION AND CLOCK
---------+---------+-----------------------------------+-----------------------------------+-------------------------
 PRN FRQ | A MJDAY |   RADORB   ALNORB   CRSORB   CLOCK |   RADORB   ALNORB   CRSORB   CLOCK | CLK_RMS CLK_CORR CLK_RESI
         |         |     (mm)     (mm)     (mm)    (mm) |    (ps)     (ps)     (ps)    (ps) |    (ps)    (ps)    (ps)
---------+---------+-----------------------------------+-----------------------------------+-------------------------
 G02   0 |   60677 |     -21        2       -4    -113 |     -71        7      -15    -376 |       0     212     -93
 G03   0 |   60677 |     -14       13        2     -73 |     -48       42        8    -243 |       0     212      17
 G04   0 |   60677 |       3       11      -19     -76 |       9       36      -62    -254 |       0     212     -51
 G05   0 |   60677 |     -31      -10       16     -73 |    -103      -33       55    -244 |       0     212      72
 G06   0 |   60677 |       6        7        8     -62 |      21       24       26    -206 |       0     212     -15
 G07   0 |   60677 |      18       13       11     -43 |      59       44       37    -144 |       0     212       9
 G08   0 |   60677 |      19      -24       14     -80 |      62      -80       45    -267 |       0     212    -116
 G09   0 |   60677 |     -19        8        8     -90 |     -63       26       27    -299 |       0     212     -24
 G10   0 |   60677 |     -10        1       10     -85 |     -32        2       32    -285 |       0     212     -40
 G11   0 |   60677 |     -11       19       20     -79 |     -37       65       68    -264 |       0     212     -14
 G12   0 |   60677 |      -1      -16      -18     -31 |      -4      -53      -61    -102 |       0     212     113
 G13   0 |   60677 |     -18      -31       -6     -72 |     -60     -103      -20    -242 |       0     212      30
 G14   0 |   60677 |       5        4      -12     -43 |      17       12      -40    -144 |       0     212      51
 G15   0 |   60677 |      -5       -9        8     -42 |     -16      -28       25    -140 |       0     212      88
 G16   0 |   60677 |      -9        4        0     -96 |     -30       14        1    -320 |       0     212     -77
 G17   0 |   60677 |     -16        0       -1     -93 |     -55        1       -3    -310 |       0     212     -44
 G18   0 |   60677 |       1        9      -10     -47 |       5       29      -34    -158 |       0     212      49
 G19   0 |   60677 |      -2       -2       18     -58 |      -8       -6       61    -194 |       0     212      26
 G20   0 |   60677 |     -25        6       11     -93 |     -84       18       36    -311 |       0     212     -15
 G21   0 |   60677 |     -23       -8        4     -85 |     -76      -25       13    -282 |       0     212       6
 G23   0 |   60677 |       8       -4        7     -45 |      26      -13       22    -150 |       0     212      36
```

Figure 4-4 Orbit and Clock Day-Boundary Discontinuities in the Alignment Log File

For the dual-frequency code bias and wide-lane day-boundary discontinuities, the columns from left to right are: satellite PRN; frequency index (0 denotes a combination formed from the two reference frequencies); Modified Julian Date (MJD); geometry-free phase bias day-boundary discontinuity (ps); reference-channel differential code bias (DCB) day-boundary discontinuity (ps); code bias day-boundary alignment correction (expected to be 0); residual reference-channel DCB discontinuity after alignment (ps); wide-lane discontinuity (milli-cycles, mcy); wide-lane day-boundary alignment correction (mcy); and residual wide-lane discontinuity after alignment (mcy).

```
2. RESULTS OF DUAL-FREQUENCY GEOMETRY-FREE BIAS ALIGNMENT
---------+---------+-------------------+-------------------+----------+-----------------
 PRN FRQ | A MJDAY |   PH_GFB    CD_GFB |   CD_CORR   CD_RESI |  WL_UPD  |  WL_CORR  WL_RESI
         |         |     (ps)      (ps) |     (ps)      (ps) |   (mcy)  |   (mcy)   (mcy)
---------+---------+-------------------+-------------------+----------+-----------------
 G02   0 |   60677 |    -191       227 |        0       227 |       14 |      -22       -7
 G03   0 |   60677 |    -105       104 |        0       104 |       26 |      -22        4
 G04   0 |   60677 |      69       -41 |        0       -41 |       21 |      -22        0
 G05   0 |   60677 |      93      -131 |        0      -131 |     -974 |      978        4
 G06   0 |   60677 |     247      -235 |        0      -235 |       21 |      -22        0
 G07   0 |   60677 |      74       -70 |        0       -70 |       26 |      -22        4
 G08   0 |   60677 |      67       -18 |        0       -18 |       17 |      -22       -4
 G09   0 |   60677 |    -128       136 |        0       136 |       20 |      -22       -1
 G10   0 |   60677 |       9        15 |        0        15 |       24 |      -22        2
 G11   0 |   60677 |     182      -171 |        0      -171 |       22 |      -22        0
 G12   0 |   60677 |     115      -170 |        0      -170 |     -979 |      978       -1
 G13   0 |   60677 |     230      -245 |        0      -245 |     -975 |      978        3
 G14   0 |   60677 |     -98        73 |        0        73 |       18 |      -22       -3
 G15   0 |   60677 |     109      -154 |        0      -154 |     -980 |      978       -1
 G16   0 |   60677 |      -9        33 |        0        33 |       12 |      -22      -10
 G17   0 |   60677 |     -16        38 |        0        38 |       20 |      -22       -1
 G18   0 |   60677 |      14       -31 |        0       -31 |       27 |      -22        5
 G19   0 |   60677 |     157      -165 |        0      -165 |       22 |      -22        1
```

Figure 4-5 Dual-Frequency Pseudorange-Bias and Wide-Lane Day-Boundary Discontinuities in the Alignment Log File

The narrow-lane phase bias day-boundary discontinuities are illustrated below. From left to right, the columns are: satellite PRN; frequency; MJD; narrow-lane phase bias day-boundary discontinuity; narrow-lane phase bias discontinuity after accounting for the orbit/clock alignment residuals; narrow-lane phase bias day-boundary alignment correction; and residual narrow-lane phase bias discontinuity after accounting for the orbit/clock alignment residuals. The final four quantities are each reported twice, once in picoseconds and once in milli-cycles.

```
3. RESULTS OF DUAL-FREQUENCY IRC ALIGNMENT

 PRN FRQ | A MJDAY |   NL_UPD      IRC |  NL_CORR IRC_RESI |   NL_UPD      IRC |  NL_CORR IRC_RESI
         |         |    (ps)     (ps) |    (ps)     (ps) |    (mcy)    (mcy) |    (mcy)    (mcy)
---------+---------+--------------------+--------------------+--------------------+-----------------
 G02   0 |   60677 |      -4      -89 |     -89        0 |      -11     -249 |     -249        0
 G03   0 |   60677 |    -253      269 |     268        1 |     -708      755 |      751        4
 G04   0 |   60677 |      33      -84 |     -89        5 |       93     -235 |     -249       14
 G05   0 |   60677 |     -14       86 |     100      -14 |      -39      241 |      281      -40
 G06   0 |   60677 |      65      -80 |     -89        8 |      183     -225 |     -249       24
 G07   0 |   60677 |      93      -85 |     -89        4 |      261     -237 |     -249       12
 G08   0 |   60677 |     -21      -95 |     -89       -7 |      -58     -267 |     -249      -19
 G09   0 |   60677 |      69      -93 |     -89       -4 |      193     -261 |     -249      -12
 G10   0 |   60677 |      47      -88 |     -89        1 |      133     -246 |     -249        2
 G11   0 |   60677 |      67      -82 |     -89        7 |      188     -229 |     -249       20
 G12   0 |   60677 |       6      107 |     100        7 |       17      301 |      281       20
 G13   0 |   60677 |     296     -266 |    -257       -9 |      830     -745 |     -719      -26
 G14   0 |   60677 |     131      -80 |     -89        9 |      366     -223 |     -249       25
 G15   0 |   60677 |     -14      102 |     100        1 |      -38      285 |      281        4
 G16   0 |   60677 |      14      -91 |     -89       -2 |       38     -254 |     -249       -5
 G17   0 |   60677 |      41      -85 |     -89        4 |      116     -238 |     -249       11
 G18   0 |   60677 |     139      -89 |     -89       -1 |      388     -250 |     -249       -2
 G19   0 |   60677 |    -250      276 |     268        8 |     -701      775 |      751       24
 G20   0 |   60677 |      76      -90 |     -89       -2 |      213     -253 |     -249       -5
 G21   0 |   60677 |     -74       80 |      79        1 |     -208      225 |      222        3
 G23   0 |   60677 |     -61       97 |     100       -3 |     -171      273 |      281       -8
 G24   0 |   60677 |     161      -76 |     -89       13 |      450     -212 |     -249       36
 G25   0 |   60677 |    -243      270 |     268        2 |     -680      757 |      751        6
```

Figure 4-6 Narrow-Lane Phase Bias Day-Boundary Discontinuities in the Alignment Log File

The code bias and phase bias alignment corrections are illustrated below. From left to right, the columns are: satellite PRN; frequency; MJD; code bias correction (ps); and phase bias correction (ps).

```
4. CORRECTIONS AND RESIDUALS FOR CODE/PHASE BIASES

4.1 CORRECTIONS AND RESIDUALS FOR PHASE BIASES
---------+---------+----------+-----------------
 PRN FRQ | A MJDAY |  PH_CORR |  PH_RESI  PH_RESI
         |         |     (ps) |     (ps)    (mcy)
---------+---------+----------+-----------------
 G02   1 |   60677 |     -110 |        0        0
 G03   1 |   60677 |      525 |       -1       -2
 G04   1 |   60677 |     -110 |       -5       -8
 G05   1 |   60677 |    -2014 |       14       22
 G06   1 |   60677 |     -110 |       -8      -13
 G07   1 |   60677 |     -110 |       -4       -7
 G08   1 |   60677 |     -110 |        7       10
 G09   1 |   60677 |     -110 |        4        7
 G10   1 |   60677 |     -110 |       -1       -1
 G11   1 |   60677 |     -110 |       -7      -11
 G12   1 |   60677 |    -2014 |       -7      -11
 G13   1 |   60677 |    -2649 |        9       15
 G14   1 |   60677 |     -110 |       -9      -14
 G15   1 |   60677 |    -2014 |       -1       -2
 G16   1 |   60677 |     -110 |        2        3
 G17   1 |   60677 |     -110 |       -4       -6
 G18   1 |   60677 |     -110 |        1        1
 G19   1 |   60677 |      525 |       -8      -13
 G20   1 |   60677 |     -110 |        2        3
 G21   1 |   60677 |     2429 |       -1       -2
 G23   1 |   60677 |    -2014 |        3        4
 G24   1 |   60677 |     -110 |      -13      -20
 G25   1 |   60677 |      525 |       -2       -3
 G26   1 |   60677 |     -744 |        4        6
 G27   1 |   60677 |    -2014 |        9       15
 G28   1 |   60677 |      525 |        2        4
```

```
4.2 CORRECTIONS AND RESIDUALS FOR CODE BIASES
---------+---------+-----+----------+-----------------
 PRN FRQ | A MJDAY | OBS |  CD_CORR |  OSB_DBD  CD_RESI  CD_RESI
         |         |     |     (ps) |     (ns)     (ps)    (mm)
---------+---------+-----+----------+-----------------
 G02   1 |   60677 | C1C |        0 |   -0.347       41       12
 G02   1 |   60677 | C1W |        0 |   -0.351       37       11
 G03   1 |   60677 | C1C |        0 |   -0.153       -7       -2
 G03   1 |   60677 | C1W |        0 |   -0.161      -15       -4
 G04   1 |   60677 | C1X |        0 |    0.062        6        2
 G04   1 |   60677 | C1C |        0 |    0.063        7        2
 G04   1 |   60677 | C1W |        0 |    0.064        8        2
 G05   1 |   60677 | C1C |        0 |    0.200      -17       -5
 G05   1 |   60677 | C1W |        0 |    0.202      -14       -4
 G06   1 |   60677 | C1C |        0 |    0.360       -7       -2
 G06   1 |   60677 | C1W |        0 |    0.364       -3       -1
 G07   1 |   60677 | C1C |        0 |    0.106      -17       -5
 G07   1 |   60677 | C1W |        0 |    0.108      -15       -5
 G08   1 |   60677 | C1C |        0 |    0.037       49       15
 G08   1 |   60677 | C1W |        0 |    0.028       40       12
 G09   1 |   60677 | C1C |        0 |   -0.210       11        3
 G09   1 |   60677 | C1W |        0 |   -0.210       12        4
 G10   1 |   60677 | C1C |        0 |   -0.019        8        2
 G10   1 |   60677 | C1W |        0 |   -0.023        4        1
 G11   1 |   60677 | C1X |        0 |    0.256      -10       -3
 G11   1 |   60677 | C1C |        0 |    0.260       -6       -2
 G11   1 |   60677 | C1W |        0 |    0.264       -2       -1
 G12   1 |   60677 | C1C |        0 |    0.263      -29       -9
 G12   1 |   60677 | C1W |        0 |    0.262      -30       -9
 G13   1 |   60677 | C1C |        0 |    0.371      -15       -4
 G13   1 |   60677 | C1W |        0 |    0.379       -7       -2
 G14   1 |   60677 | C1X |        0 |   -0.118      -17       -5
 G14   1 |   60677 | C1C |        0 |   -0.118      -17       -5
 G14   1 |   60677 | C1W |        0 |   -0.113      -11       -3
 G15   1 |   60677 | C1C |        0 |    0.234      -22       -7
 G15   1 |   60677 | C1W |        0 |    0.238      -18       -5
 G16   1 |   60677 | C1C |        0 |   -0.050       42       12
 G16   1 |   60677 | C1W |        0 |   -0.051       40       12
 G17   1 |   60677 | C1C |        0 |   -0.056       12        3
 G17   1 |   60677 | C1W |        0 |   -0.058       10        3
```

（a）Corrections and Residuals of Code bias （b）Corrections and Residuals of Phase bias

Figure 4-7 Corrections Applied to Observable Biases in the Alignment Log File

Precise Orbit, Clock and Bias Product Files

The precise product files cover at least two consecutive days. The original input products remain unchanged. The target-day clock and bias products are aligned to the corresponding products of the preceding day. When orbit or clock extrapolation is performed, a corresponding "_predicted" SP3 or CLK file is generated with the extrapolated records appended after the last original epoch. When Write DOCB is set to YES, the precise bias files contain DOCB records, which quantify the residual discontinuities at day boundaries. The format is illustrated below.

```
+SOLUTION/DAY_BOUNDARY_DISCONTINUITY
*DBD  SVN_ PRN STATION__ OBS1 OBS2 MIDNIGHT_AT___ MIDNIGHT_AT___ UNIT __ESTIMATED_VALUE____ _STD_DEV___
 DOCB G061 G02           C1C       2025:001:86400 2025:002:00000 ns                0.04127     0.00000
 DOCB G061 G02           C1W       2025:001:86400 2025:002:00000 ns                0.03707     0.00000
 DOCB G061 G02           C2W       2025:001:86400 2025:002:00000 ns                0.00122     0.00000
 DOCB G061 G02           L1C       2025:001:86400 2025:002:00000 ns               -0.00009     0.00000
 DOCB G061 G02           L1W       2025:001:86400 2025:002:00000 ns               -0.00009     0.00000
 DOCB G061 G02           L2W       2025:001:86400 2025:002:00000 ns               -0.00009     0.00000
 DOCB G069 G03           C1C       2025:001:86400 2025:002:00000 ns               -0.00697     0.00000
 DOCB G069 G03           C1W       2025:001:86400 2025:002:00000 ns               -0.01487     0.00000
 DOCB G069 G03           C2X       2025:001:86400 2025:002:00000 ns                0.01083     0.00000
 DOCB G069 G03           C2S       2025:001:86400 2025:002:00000 ns               -0.02337     0.00000
 DOCB G069 G03           C2L       2025:001:86400 2025:002:00000 ns               -0.01467     0.00000
 DOCB G069 G03           C2W       2025:001:86400 2025:002:00000 ns               -0.01357     0.00000
 DOCB G069 G03           L1C       2025:001:86400 2025:002:00000 ns               -0.00148     0.00000
 DOCB G069 G03           L1W       2025:001:86400 2025:002:00000 ns               -0.00148     0.00000
 DOCB G069 G03           L2X       2025:001:86400 2025:002:00000 ns               -0.00148     0.00000
 DOCB G069 G03           L2S       2025:001:86400 2025:002:00000 ns               -0.00148     0.00000
```

Figure 4-8 DOCB Output Format

4.3.2 Usage of result data processing scripts/programs

Note

As shown in Table 4-2, there are some useful scripts under the "scripts/" directory that help users process. Note that if your Python 3 path is not "/usr/bin/python3", you can use the ***which*** command to find the path of python3 in your computer.

```
which python3
/usr/local/bin/python3
```

Then, modify these python programs' header to the corresponding path.

```
#!/usr/bin/env python3
#!/usr/local/bin/python3
```

Copy the modified python scripts to the path where software environment variables are located (~/.PRIDE_DBAlign_BIN/). Then you can use these python scripts directly in any directory.

```
cp plot-*.py ~/.PRIDE_DBAlign_BIN/
```

Moreover, the Python plotting scripts require the Matplotlib packages.

Usage of ***pdba_plot***

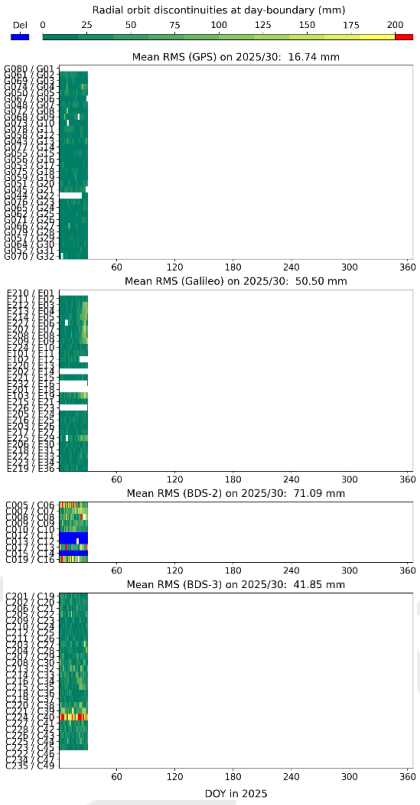
pdba_plot is used to generate heat maps of the residual day-boundary discontinuities after product alignment. Its usage is:

```
pdba_plot aligned_filefolder
```

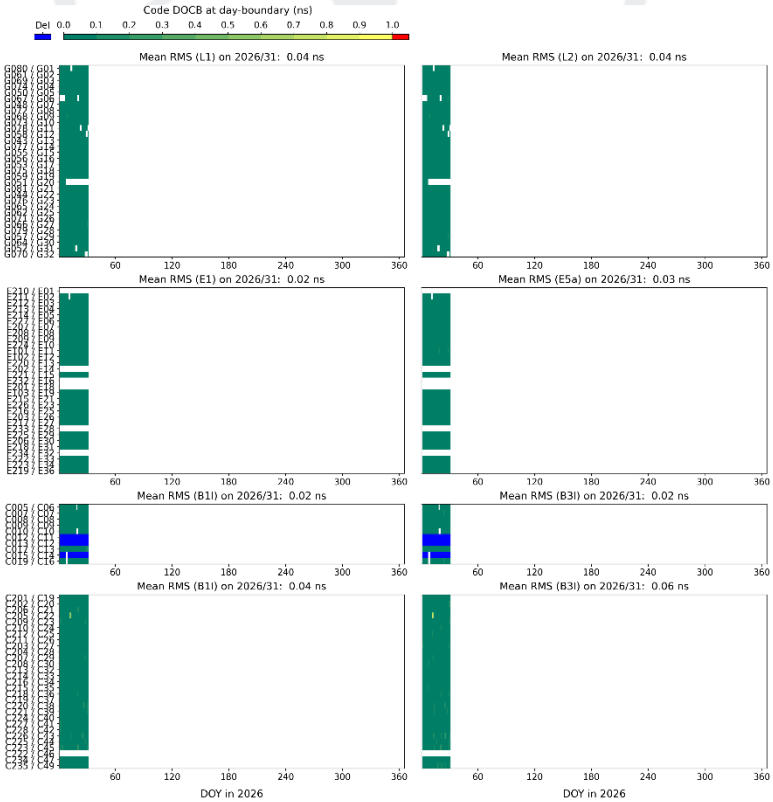
where

```
aligned_filefolder      Directory containing the logs and products for the
                        aligned day boundaries.
```

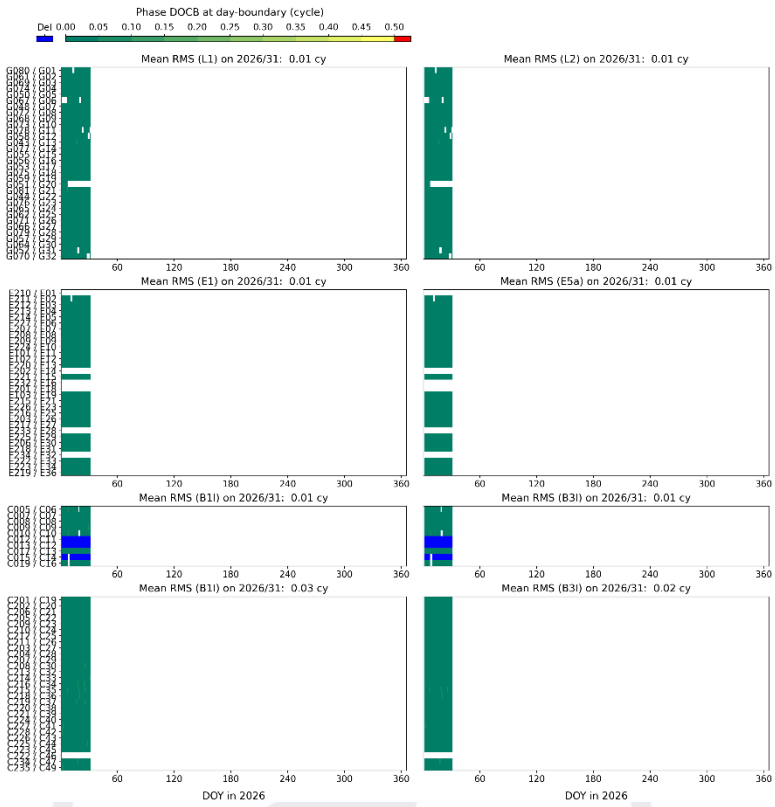
An example of the pdba_plot output is shown below, using the alignment logs for WUM Rapid products from day of year (DOY) 001 to 031 in 2026.



(a) Radial-orbit day-boundary discontinuities



(b) Code DOCBs at day-boundary



(c) Phase DOCB at day-boundary

Figure 4-9 Example Output from pdba_plot

4.4 Quick Start and Program Execution

After the software installation is completed, the executable programs, processing scripts, configuration template, plotting programs, and satellite parameter file are installed in the directory ~/.PRIDE_DBAlign_BIN/. The installer also adds this directory to the system PATH. On Linux, the PATH setting is written to ~/.bashrc; on macOS, it is written to ~/.bash_profile. Users should open a new terminal or reload the corresponding shell configuration file before running the software.

PRIDE DBAlign uses algirc as its core day-boundary alignment program. Although algirc is driven by a configuration file, the pdba command provides a simplified command-line interface. It automatically determines the processing date from the input filenames, locates the corresponding preceding day products, generates an ALGIRC configuration file, and invokes algirc to perform the alignment.

In this section, the software installation directory is denoted by path-to-software, the project working directory by path-to-project, the configuration file by config_file. All commands below are executed in path-to-project.

4.4.1 Usage of *pdba*

The usage of ***pdba*** is shown in Figure 4-10. For more detailed information, you can enter ***pdba*** in the terminal.

```
Usage: pdba -clk <file> -bia <file> -sp3 <file> [options]

Align the target-day clock and bias products to the previous day's original
products using the corresponding SP3 orbits. The target date is read from
the product filenames.

Start up:
  -V, -v, --version                       display version information
  -H, -h, --help                          display this help

Product options:
  -clk <file>, --clock <file>             target-day clock product (required)
  -bia <file>, --bias <file>              target-day code/phase bias product (required)
  -sp3 <file>, --sp3 <file>               target-day SP3 orbit product (required)

Processing options:
  -cfg <file>, --config <file>            configuration file for PRIDE DBAlign
  -wdocb <YES|NO>                         write day-boundary discontinuity records
                                            * default: YES
  -sys <char>, --system <char>            GNSS systems selected from G, E and C
                                            * default: GEC)
  -ubd23 <YES|NO>, --unify-bds23 <YES|NO>  unify BDS-2/3 datum
                                            * default: YES for MJD > 61152, otherwise NO

Configuration precedence:
  command-line option > configuration file > built-in default
```

Figure 4-10 pdba script partial help information

Except for "**-clk**", '**-sp3**' and "**-bia**", all other options are optional. The order of precedence is Command Line Parameters > Configuration File > Built in Default Value. The "**-clk**", '**-sp3**' and "**-bia**" options must always be specified on the command line and are never read from the configuration file. Option names must be entered exactly as defined, while the values of "**-wdocb**", "**-sys**" and "**-ubd23**" may be entered in either uppercase or lowercase. In the command syntax, < > indicates a required parameter and [ ] indicates an optional parameter. The options can be divided into three categories: (1) Mandatory product options; (2) Optional product and processing options; and (3) Start-up options. The meaning of each option is described below:

(1) Mandatory product options:

- **-clk <file>, --clock <file>**: Target day satellite clock product. This option is required.
- **-bia <file>, --bias <file>**: Target day code/phase bias product. This option is required.
- **-sp3 <file>, --sp3 <file>**: Target day satellite orbit product. This option is required.

(2) Optional product and processing options:

- **-cfg <file>, --config <file>**: Optional base ALGIRC configuration file. Either a relative or an absolute path may be used. A relative path is interpreted with respect to the current working directory. If omitted, the default configuration installed with ***pdba*** is used.
- **-sys <char>, --system <char>**: GNSS systems to be processed. One or more characters may be selected from G, E, and C, corresponding to GPS, Galileo, and BDS. For example, -sys GE processes GPS and Galileo. When neither -cfg nor -sys is specified, the default is GEC.
- **-ubd23 <YES|NO>, --unify-bds23 <YES|NO>**: Controls whether the BDS-2 and BDS-3 clock datums are unified. The value is case-insensitive. When neither -cfg nor -ubd23 is specified, the default is NO for MJD earlier than 61152 and YES otherwise. MJD 61152 (April 22, 2026) marks the completion of the BDS reconfiguration considered here.
- **-wdocb <YES|NO>**: Controls whether DOCB records are written to the BIA file. The value is case-insensitive. The default is YES.

(3) Start-up options:

- **-V, -v, --version**: Displays the version information of pdba.
- **-H, -h, --help**: Displays the command-line help information.

Note: pdba processes only one day boundary in each invocation and performs forward alignment only. For a target day D, the original clock, orbit, and bias products for both day D and day D−1

must be placed in the same directory. This requirement also applies when the two days cross a calendar-year boundary. ***pdba*** determines the processing date from the product filenames. For multi-day processing, run ***pdba*** separately for each target day.

4.4.2 Configuration File

The configuration template is located at table/config_template and uses the "Keyword = Value" format; text following "!" is treated as a comment, while lines beginning with # or * are treated as comment lines. The satellite list is defined as a separate block and does not use the Keyword = Value format.

In the current version of ***pdba***, product files are specified primarily through command-line options rather than filename templates in the base configuration. The target day clock, orbit and bias products must be supplied using -clk, -sp3 and -bia, respectively. pdba extracts the date from the target filenames, changes only the detected date field to construct the expected preceding day filenames, and searches for those files in the same product directory. It then generates a concrete configuration named config_pdba_YYYYDDD in that directory.

During this process, pdba overwrites Product directory, Satellite orbit, Satellite clock, Code/phase bias. The base configuration is mainly used to provide Write DOCB, Unify BDS-2/3, and the satellite list, subject to command-line options.

Product Paths and File Names

- Product directory = <product directory>

pdba derives the product directory from the target clock file. The target day clock, bias, and orbit products must be located in the same directory. The corresponding preceding day products must also be present in that directory. The generated configuration, ALGIRC log, and aligned products are written there.

- Satellite orbit = <preceding day SP3> <target day SP3>

The target day orbit file is supplied using -sp3. ***pdba*** constructs and locates the preceding day filename by replacing the detected date field in the target filename.

- Satellite clock = <preceding day CLK> <target day CLK>

The target day clock file is supplied using -clk. pdba constructs and locates the preceding day filename by replacing the detected date field in the target filename.

- Code/phase bias = <preceding day BIA> <target day BIA>

The target day OSB file is supplied using -bia. The corresponding preceding day file is located using the same date-field replacement rule as for the clock product.

- ERP file = <preceding day ERP>

The preceding day ERP file is used for coordinate transformations when the preceding day orbit product lacks the 24:00 epoch and orbit extrapolation is required. When -cfg is not specified, ***pdba*** automatically searches for this file in the preceding day SP3 directory. If no ERP file is available, the simplified coordinate transformation is used instead.

Alignment Options and Satellite List

- Write DOCB = YES

Write DOCB controls whether DOCB (Discontinuity of Orbits, Clocks, and Biases) records are written to the output BIA products. It should normally remain set to YES to support subsequent quality checks.

- Unify BDS-2/3 = Default

  Unify BDS-2/3 specifies whether BDS-2 and BDS-3 are treated as a unified system. The default is NO for MJD earlier than 61152 and YES otherwise.

- +GNSS satellites / *PRN / -GNSS satellites

This section defines the satellites included in the processing. Prefixing an individual satellite line with # disables that satellite.

4.4.3 General Operation Steps

After installation, consecutive daily products can be aligned as follows:

1. Confirm that install.sh has been executed successfully and that pdba can be invoked directly in a new terminal session.
2. Prepare uncompressed CLK, SP3, and BIA products for two consecutive days in the same product directory. Both the preceding day and target day products must be available.
3. Run pdba for the target day. For example:

```
pdba \
  -clk WUM0MGXRAP_20260020000_01D_30S_CLK.CLK \
  -bia WUM0MGXRAP_20260020000_01D_01D_OSB.BIA \
  -sp3 WUM0MGXRAP_20260020000_01D_05M_ORB.SP3 \
  -ubd23 NO
```

For multi-day processing, run pdba separately for each target day.

4. Inspect the product directory. It contains the generated config_pdba_YYYYDDD configuration, the algirc-MJD.out log, and the aligned clock and bias products (and any "_predicted" SP3 or CLK files generated by extrapolation).
5. To examine the residual discontinuities at day boundaries, run:

```
pdba_plot path-to-products
```

pdba_plot requires Python and Matplotlib. It generates algirc-sum.out and the radial orbit, Code and Phase DOCBs plots in the specified directory.

4.4.4 Processing Example

The example/wum0rap directory provides examples based on WUM Rapid products. It contains SP3, CLK, and BIA products for DOY 001 and 002 of both 2025 and 2026. Because each alignment requires the preceding day products, DOY 002 is aligned using DOY 001 as the reference day.

Example: Day-Boundary Alignment of WUM Rapid Products

1. Change to the example directory:

```
cd path-to-software/example
```

2. Confirm that the required products are available in:

   wum0rap/2025

   wum0rap/2026

3. Run the example script:

```
./test.sh
```

The script performs the DOY 002 alignment for both 2025 and 2026 using the -clk, -bia, -sp3, and -ubd23 options.

4. Inspect the generated configurations, ALGIRC logs, and aligned products in the corresponding

year directories. The results can be plotted using:

```
pdba_plot wum0rap/2025
pdba_plot wum0rap/2026
```

To process products from another analysis center, provide the corresponding target day CLK, BIA, and SP3 filenames. The filenames must contain a supported date field, and the preceding day products must follow the same naming convention and be located in the same directory.

Appendix A. Instruction on Day-Boundary Aligned Products

This guide is intended for PPP users who want to apply aligned products (including orbits, clocks, and biases) to PPP software other than PRIDE PPP-AR. Users must be able to modify their software accordingly.

Following day-boundary alignment of the precise satellite products, clock estimates at 24:00:00 on the preceding day (the same boundary epoch as 00:00:00 on the target day) are provided for each GPS, Galileo, and BDS satellite in the CLK files. In addition, a numerical continuity indicator at this epoch, referred to as the DOCB (Discontinuity of Orbits, Clocks and Biases), is defined and provided in the Bias-SINEX files. To ensure optimal continuous data processing, users' PPP software must:

Read the estimates for the 24:00:00 epoch from the preceding day's product and the 00:00:00 epoch from the target day's product. Prevent orbit and clock interpolations across day boundaries (i.e., 24:00:00 or 00:00:00). This mechanism is illustrated in the following diagram. Taking orbit interpolation as an example, conventional methods (left) tend to place the interpolation point in the middle of the interpolation interval. However, in our proposed method (right), when the interpolation point falls on the preceding day, the epoch 24:00:00 of the preceding day's products should be used; when the interpolation point falls on the target day, the epoch 00:00:00 of the target day's products should be used instead. Because the interpolation methods for orbit and clock differ, the compatibility of their interpolation results will be compromised by the discontinuities in orbits and clocks. However, this problem can be avoided by interpolating them within a day.

